

Grade 8
Unit 1
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

For questions 1-5, evaluate each expression using the order of operations.

$$\begin{aligned} 1. \quad | - 5.2 | + (5 \cdot 2.1)^2 + | 6.75 | & \quad 5.2 + (10.5)^2 + 6.75 \\ & \quad 5.2 + 110.25 + 6.75 \\ & \quad 122.2 \end{aligned}$$

$$\begin{aligned} 2. \quad -2.6|(-1)^8 + 9.72| & \quad -2.6|1 + 9.72| \\ & \quad -2.6|10.72| \\ & \quad -2.6 \cdot 10.72 \\ & \quad -27.872 \end{aligned}$$

$$\begin{aligned} 3. \quad (7.9 + (-1.5))^2 \div (-8) \times 3 & \quad (6.4)^2 \div (-8) \times 3 \\ & \quad 40.96 \div (-8) \times 3 \\ & \quad -5.12 \times 3 \\ & \quad -15.36 \end{aligned}$$

$$\begin{aligned} 4. \quad \left(-\frac{-3-5}{2}\right)^3 \cdot (6.2) & \quad \left(-\frac{-8}{2}\right)^3 \cdot (6.2) \\ & \quad (4)^3 \cdot 6.2 \\ & \quad 64 \cdot 6.2 \\ & \quad 396.8 \end{aligned}$$

$$\begin{aligned} 5. \quad 8\frac{3}{4} - 16.4 \div 4 - | - 5.8 | & \quad 8\frac{3}{4} - 4.1 - 5.8 \\ & \quad 4.65 - 5.8 \\ & \quad -1.15 \end{aligned}$$

Tyler simplified the expression below and made a few errors.

$$\begin{aligned} &1\frac{1}{2} - 2(2.5^2) - |-6| \\ &1\frac{1}{2} - 2(6.25) - |-6| \\ &-0.5(6.25) - |-6| \\ &-3.125 - |-6| \\ &-3.125 + 6 \\ &2.875 \end{aligned}$$

6. Circle one of the mistakes and then explain the mistake below.

Tyler subtracted before multiplying $2 \cdot 6.25$.

7. Circle a second mistake and then explain the mistake below.

Tyler did not correctly evaluate the absolute value, $|-6|$.

8. Simplify the expression correctly.

$$1\frac{1}{2} - 2(2.5)^2 - |-6|$$

$$1\frac{1}{2} - 2(6.25) - |-6|$$

$$1\frac{1}{2} - 12.5 - 6$$

$$-11 - 6$$

$$-17$$

9. True or False? *Always perform the order of operations from left to right.*

True

10. True or False? *Perform operations with exponents first when evaluating using the order of operations.*

False, parenthesis (groupings) should be completed before exponents.